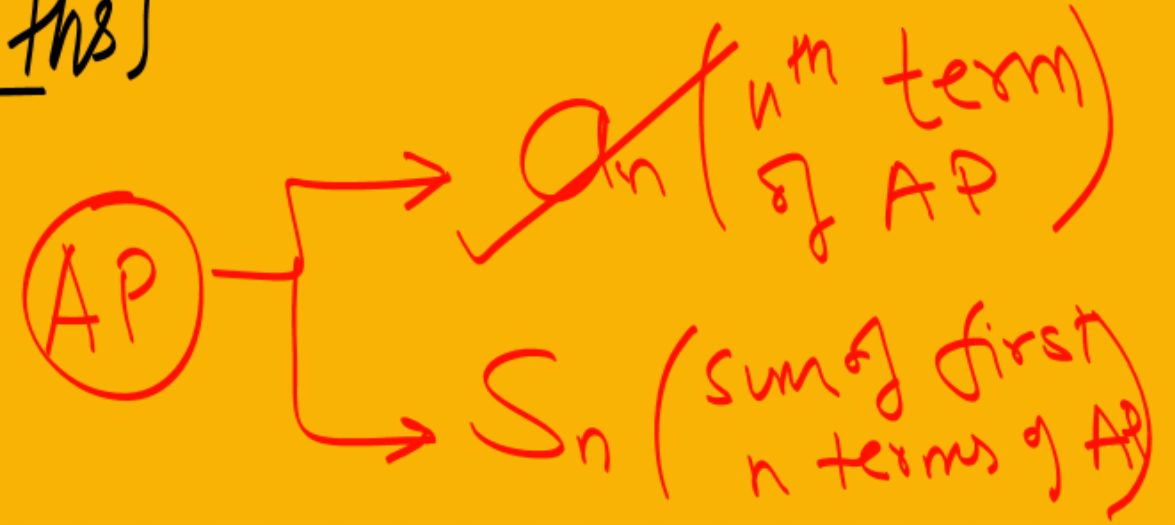


[Aim: 100/100 in Maths]

आरंभ



ARITHMETIC PROGRESSION

Lecture #7

Sum of First 'n' terms of an A.P.:

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_n = \frac{n}{2} [a + \boxed{a + (n-1)d}]$$

$$S_n = \frac{n}{2} [a + a_n]$$

a_n

किसी AP में

अंतिम last term
दी हो

$$S_n = \frac{n}{2} [a + l]$$

$l = \text{last term of AP.}$

#LP: Find the sum of first 30 terms of an A.P. whose second term is 2 and seventh term is 22.

$$a_2 = 2 \quad | \quad a_7 = 22$$
$$\boxed{a + d = 2} \text{---(I)} \quad \boxed{a + 6d = 22} \text{---(II)}$$

$$\boxed{a = -2}$$
$$\boxed{d = 4}$$

$$\begin{matrix} a_n \checkmark \\ S_n \checkmark \end{matrix}$$

put

To find

$$S_{30} = \frac{30}{2} [2a + (30-1)d]$$

$$S_{30} = \frac{30}{2} [2(-2) + 29(4)]$$

$$S_{30} = \frac{30}{2} [-4 + 116]$$

$$= \frac{30}{2} \times 112$$

$$\boxed{S_{30} = 1680}$$

**#LP: The sum of first six terms of an arithmetic progression is 42.
The ratio of its 10th term to its 30th term is 1:3. Calculate the first
and the thirteenth term of the A.P. [CBSE 2009]**

$$S_6 = 42$$

$$3 \times \frac{6}{2} [2a + (6-1)d] = 42$$

$$3 [2a + 5d] = 42 \quad (\because d=a)$$

$$2a = 14$$

$$a = 2 \quad \checkmark$$

$$d = 2 \quad \checkmark$$

$$\frac{a_{10}}{a_{30}} = \frac{1}{3}$$

$$\Rightarrow \frac{a+9d}{a+29d} = \frac{1}{3}$$

$$\rightarrow 3a+27d = a+29d$$

$$\rightarrow 3a - a = 29d - 27d$$

$$\rightarrow 2a = 2d$$

$$a = d \quad \text{--- (1)}$$

To find:

$$a = ?$$

$$a_1 = 2 \quad \checkmark$$

$$a_{13} = a + 12d$$

$$= 2 + 12(2)$$

$$= 2 + 24$$

$$a_{13} = 26 \quad \checkmark$$

#LP: Find the sum of first n natural numbers.

$$1 + 2 + 3 + 4 - - - + n \leftarrow l$$

$$S_n = \frac{n}{2}(a+l)$$

$$S_n = \frac{n}{2}(1+n)$$

Sum of first
n natural
number

$$S_n = \frac{n(n+1)}{2}$$

$$d = 51 - 54 = -3$$

#LP: How many terms of the series 54, 51, 48, ... be taken so that their sum is 513? Explain the double answer. [CBSE 2005]

① → check AP?? Yes ✓

② let 'n' terms are there

$$S_n = 513$$

$$\Rightarrow \frac{n}{2} [2(54) + (n-1)(-3)] = 513$$

$$\Rightarrow n [108 - 3n + 3] = 513 \times 2$$

$$\Rightarrow n (111 - 3n) = 1026$$

$$\Rightarrow 111n - 3n^2 = 1026$$

$$3n^2 - 111n + 1026 = 0$$

$$3[n^2 - 37n + 342] = 0$$

$$n^2 - 37n + 342 = 0$$

$$n = 18, 19$$

$$\left(\begin{array}{l} S_{18} = 513 \\ S_{19} = 513 \end{array} \right) \text{ Ans}$$

$$\begin{array}{l} 1 \longrightarrow 18 \Rightarrow 513 \\ 1 \longrightarrow 19 = \underline{\underline{513}} \end{array}$$

$$d = -ve$$

only possible when Q_{19} is 0.

checking

$$\begin{aligned} Q_{19} &= a + 18d \\ &= 54 + 18(-3) \\ &= 54 - 54 \end{aligned}$$

$$Q_{19} = 0$$

← this is the reason of two answers.

#LP: If the nth term of an A.P. is $(2n + 1)$, find the sum of first n terms of the A.P. [CBSE 2005]

$$a_n = 2n + 1$$

$$n=1 \rightarrow a_1 = 2(1) + 1 \Rightarrow \textcircled{3} = a_1$$

$$n=2 \rightarrow a_2 = 2(2) + 1 \Rightarrow 5 = a_2$$

$$\begin{aligned} d &= a_2 - a_1 \\ &= 5 - 3 \Rightarrow \textcircled{2} \end{aligned}$$

To find

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$\downarrow$ $a=3$ $\downarrow$ $d=2$

$$S_n = \frac{n}{2} [2(3) + (n-1)(2)]$$

$$S_n = \frac{n}{2} [6 + 2n - 2]$$

$$S_n = \frac{n}{2} [2n + 4]$$

$$S_n = \frac{n \times 2(n+2)}{2}$$

$$\boxed{S_n = n(n+2)} \quad \#$$

#LP: Solve the equation: $1 + 4 + 7 + 10 + \dots + x = 287$.

① check AP ✓

let 'n' terms

$$\therefore a_n = x$$

$$a + (n-1)d = x$$

$$1 + (n-1)(3) = x$$

$$1 + 3n - 3 = x$$

$$3n - 2 = x$$

$$3n = x + 2$$

$$n = \frac{x+2}{3}$$

a_n

$$\frac{n}{2}(a+l) = 287$$

$$\frac{n}{2}(1+x) = 287$$

$$\frac{x+2}{3 \times 2}(1+x) = 287$$

$$(x+2)(x+1) = 287 \times 6$$

$$x = ??$$

THANK YOU COODIESS !!

